ECE 447

Lesson 20
PCM (cont'd) and Pulse Modulations



Schedule and Admin

- Schedule
- Admin
  - **HW4**. Due Lesson 21 (Tue, 6 Oct) to Gradescope.
  - **HW5**. Assigned Lesson 21 (Tue, 6 Oct); due Lesson 23 (Tue, 13 Oct).
  - **Lab 5**. Begins Lesson 21 (Tue, 6 Oct); due Lesson 24 (Thu, 15 Oct).

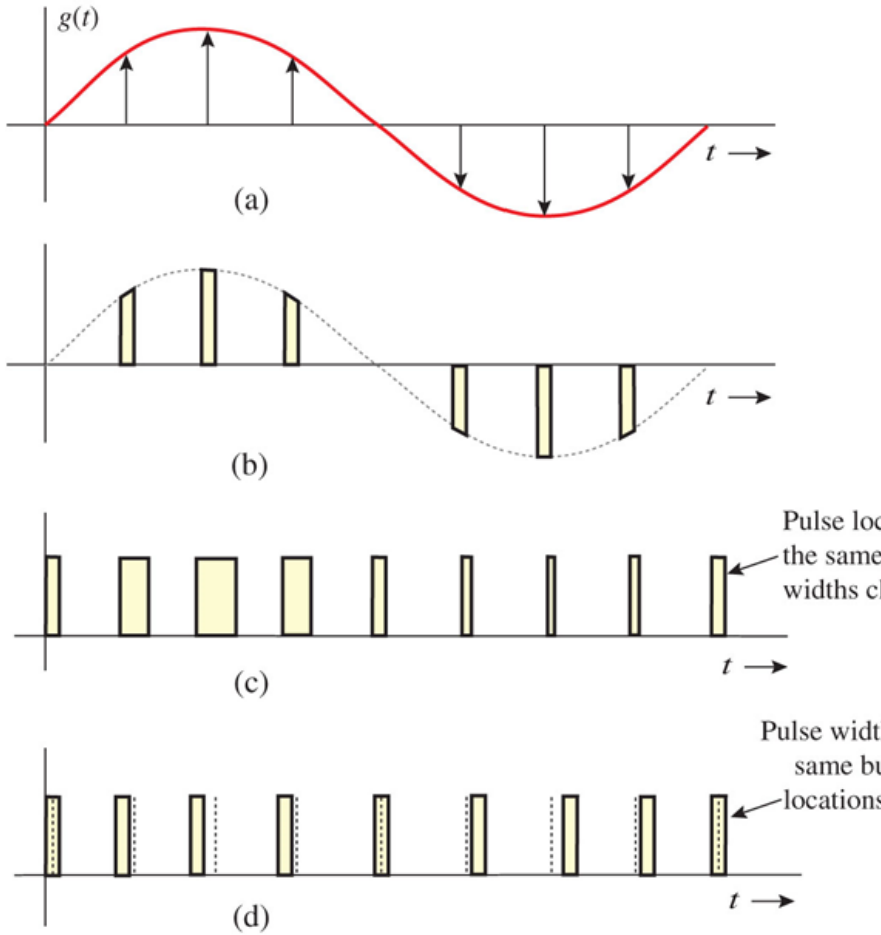
Binary PCM

- SNR
  - Rule of thumb: each additional bit adds 6dB of SNR (how many times greater is that?)
- Max information rate given finite bandwidth
  - Assuming a noiseless channel with bandwidth B, a max of $2B$ pieces of info (e.g., bits) per second can be transmitted error-free
  - Shannon-Hartley theorem defines limit in presence of noise
- Transmission Bandwidth
  - For $m(t)$ band-limited to BHz, min sample rate is $2B$ samples/s
  - Encoder output is then $2nB$ bit/s
  - Minimum required (theoretical) channel bandwidth,

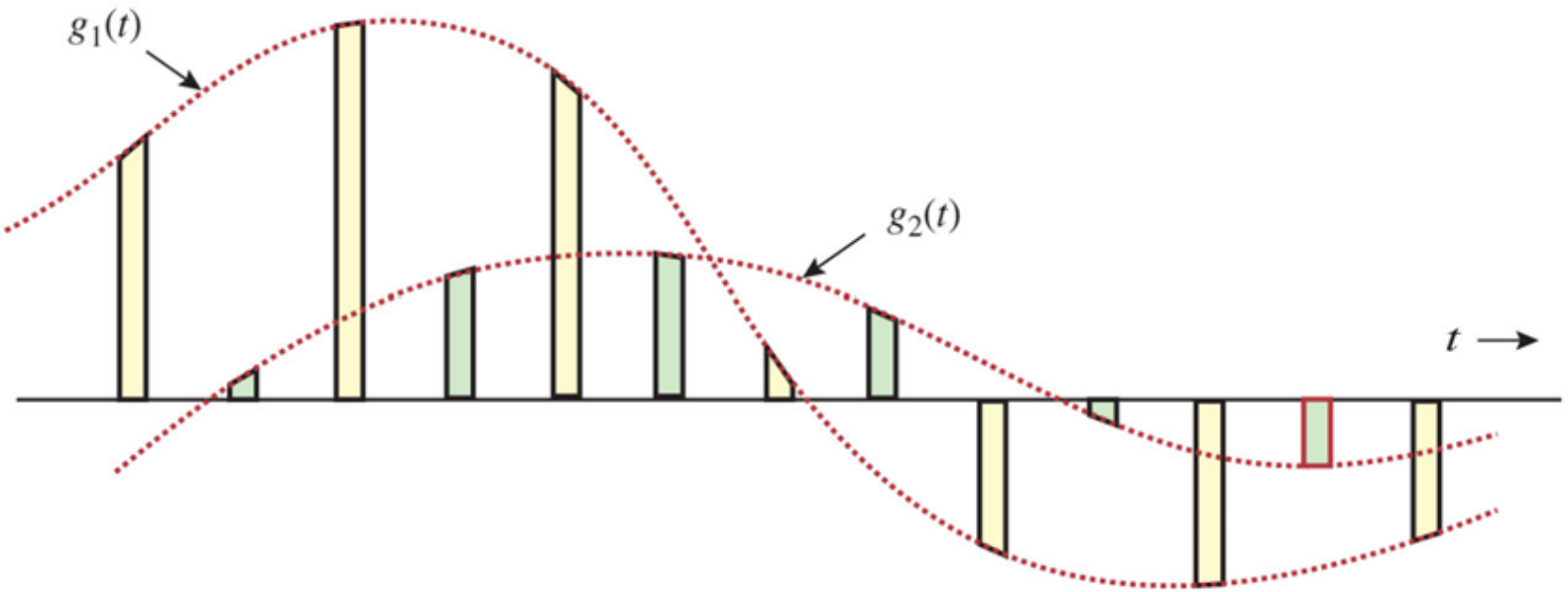
$$B_T =$$

Pulse Modulations

- **PAM**: primarily now used for baseband digital communications (e.g., USB4 Ver 2.0 uses PAM-3, or 3-level PAM)
- **PWM**: power or motor control (e.g., robotics, power supply regulation); minimal telecommunication applications
- **PPM**: fiber optical communications, deep-space communications, R/C applications



Time Division Multiplexing



Binary PCM - Example Problem 5.2

- $m(t)$ band-limited to 3kHz is sampled at 1/3 higher than the Nyquist rate
- Max acceptable quantization error is 0.5% m_p
- Find min channel bandwidth for transmission.
- If 24 signals combined via TDM, what is min required transmission bandwidth of multiplexed signal?